Freight Rate Prediction

Data

I worked on 48,000 loads from January 1 to October 31, 2025. Each load has the pickup and delivery cities and their coordinates, the distance, the equipment type, the weight, the date, two columns called market_index and quote_signal, and the price we want to predict.

Data issues I found and how I handled them

Negative weights: 292 loads have a negative weight. When I compared these values without the sign to the normal weights, they had almost exactly the same distribution, the same smallest and largest value and the same mean. So I assumed they are real weights that were written with a minus sign by mistake and fixed them. The same problem exists in the validation file and I fixed it there too.

Wrong prices: The price alone is not enough to judge, so I calculated the rate per mile. Most loads are between 2 and 2.3 dollars per mile, but there are two groups far from the rest with a clear gap between them: 337 loads below 1.4 dollars per mile and 340 loads above 4 dollars per mile. These are 677 loads, about 1.4% of the data. I removed them from training because there is no way to know their real price, and if they stay the model will learn wrong things from them.

quote_signal: I noticed its values are in the same range as the rate per mile. When I checked, in months 1, 2, 3, 6 and 9 it is the same as the rate per mile in about 98% of the loads, and in the other months it matches in only about 10%. So it leaks the answer in some months and is random in the rest. We can't rely on it for the future, so I dropped it.

Validation file: It is all from November and December, months the model has never seen. It also has 8 cities that are not in the training data, so I used the coordinates instead of the city names, so the model can handle a new city from its location.

December file: It has no market_index, no quote_signal and no coordinates. I got the coordinates back from the city names, because I checked that each city has one fixed set of coordinates across all the data.

Things I checked that turned out fine: no duplicated rows, no distances that are negative or zero, and no loads from a city to itself. The equipment type has only 3 values, all written correctly. The same route can have a slightly different distance, within about 3%, which is normal.

Split and validation approach

Since the task is to predict upcoming months, I split the data by time, not randomly: I trained on January to August (37,944 loads) and tested on September and October (9,379 loads). This way the model is tested on months it hasn't seen, exactly like the real case. If I had split randomly, the model would see loads from the same days I test on, and the result would look better than reality.

I measured the error with two metrics: MAE, the average error in dollars, and MAPE, the average error as a percentage of the price.

Before the model I built a simple baseline: price = median rate per mile for the equipment type times the distance. This gives me a ruler to know if the model really adds something.

Model	MAE	MAPE
Baseline	$175	8.1%
Model without market_index	$51	2.2%
Model with market_index	$60	2.5%

Model

I used HistGradientBoostingRegressor from scikit-learn. I chose it because it is strong with tabular data, handles missing values by itself (the missing weights), and is fast.

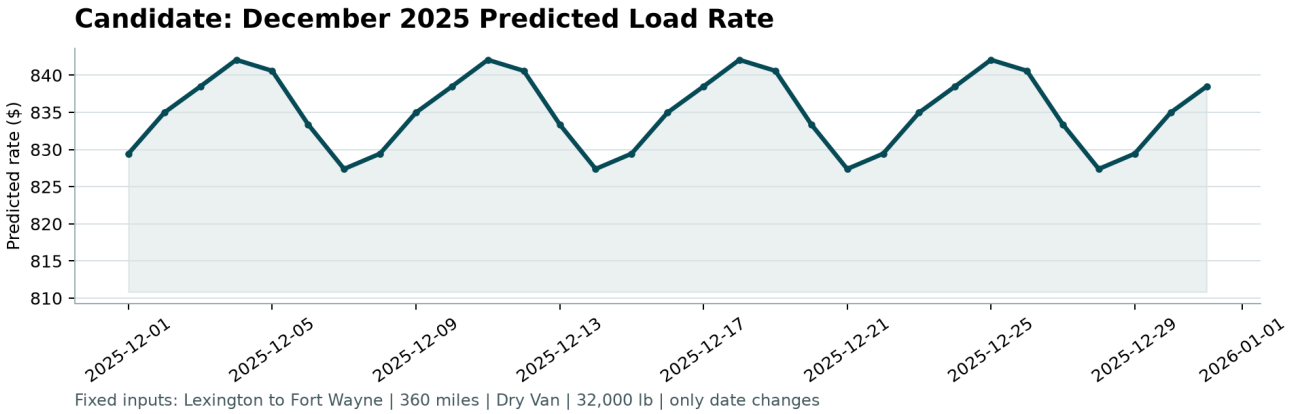
I made the model predict the rate per mile instead of the price itself, because it is an almost constant number and easier to learn, and then I multiply it by the distance.

Features: distance, weight, coordinates of both cities, equipment type, month and day of week.

For market_index I didn't want to decide from my head, so I tried the model with it and without it. With it the error got bigger, and it is also missing in the December file, so I dropped it and kept one model that works on all files.

Finally I trained the model on all the clean data (January to October) before predicting November and December, so it learns from the latest months.

December chart



The prices are between 827 and 842 dollars, about 2.3 dollars per mile. The line has a repeating weekly pattern: mid week is a bit more expensive than the weekend, which is the same as what I saw in the data (about 2% difference).

Limitation: the model has not seen a single day of December in training, so it doesn't know about holiday seasons like Christmas. That's why the chart repeats the same week.